

Data Wrangling 🤠

Tools Tour

Lesson 1

Visual Studio Code

Three areas you'll use every day:

- **Explorer** (left sidebar) — browse and open files in your project
- **Editor** (center) — where you read and write code
- **Terminal** (bottom panel) — where you run code

The Terminal & Shell

- The **terminal** is a text-based interface for talking to your computer
- The **shell** is the program inside the terminal that reads your commands and carries them out
- Instead of clicking buttons, you type instructions

The Python REPL

- **REPL** stands for Read-Eval-Print Loop
- An interactive Python session — you type code, Python runs it immediately
- Great for experimenting and testing ideas quickly
- Nothing is saved when you close it

Demo: Opening the REPL

1. Open up **Visual Studio Code** with your `data_wrangling` folder
2. Open up a **Terminal** by clicking the Settings Gear → Command Palette → Search for "Terminal: Create New Terminal"
3. Type the following command and press Enter:
 - **Mac:** `python3`
 - **Windows:** `python`
4. You should see the `>>>` prompt — that's the REPL!
5. Play around. Once you `quit()` or close the terminal, your commands and work will be lost.

Demo: REPL

```
>>> 2 + 2
4
>>> 10 / 3
3.3333333333333335
>>> "Data" + " " + "Wrangling"
'Data Wrangling'
>>> 2 ** 8
256
```

Python Programs

- A **program** is code saved in a file (`.py` extension)
- Unlike the REPL, programs can be saved and run again
- We run programs from the terminal

Demo: Running a Program

Open `lessons/tour.py` in the editor. It contains:

```
# tour.py  
print("I am touring VS Code.")
```

Run it from the terminal:

- **Mac:** `python3 -m lessons.tour`
- **Windows:** `python -m lessons.hello`

You should see: `I am touring VS Code.`

REPL for quick experiments.
Programs to save and reuse your work.